

SHREYA CHANDRASEKARAN

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EDUCATION

University of Toronto

Bachelor of Applied Science, Mechanical Engineering Studies

Sep 2023 – Apr 2025

Ontario, Canada

GEMS Wellington School

A-Levels: Mathematics (A), Physics (B), Chemistry (B), Design & Technology (C)

Sep 2018 – Jun 2023

Doha, Qatar

MAJOR PROJECTS

Camera Clip | Design Engineer

Jan 2025 – May 2025 | University of Toronto

- Designed a quick-access camera carrying solution enabling single-hand attach/detach in < 2s, meeting ergonomics, security, and durability specs
- Explored a broad design space using structured ideation and reverse engineering; down-selected concepts using feasibility analysis and weighted decision matrices
- Modeled in Fusion 360 and iteratively developed FDM prototypes; validated design through load and dimensional testing and produced engineering documentation

Fall Detection Device (Biomedical) | Project Manager

Jan 2024 – May 2024 | University of Toronto

- Redesigned fall-detection device for long-term care facilities, establishing performance targets of $\geq 95\%$ accuracy and $\leq 5\%$ false positives
- Produced engineering sketches, Gantt charts, and development milestones to structure the design and testing process
- Integrated accelerometer and gyroscope; optimized casing, sensor placement, and sampling frequency for improved signal reliability
- Tested preliminary prototype in controlled settings to assess performance against defined metrics

Aluminum Can Design Analysis | Team member

Mar 2024 | University of Toronto

- Modeled can geometry as a cylindrical system; applied constrained optimization to minimize surface area for a fixed volume
- Analyzed structural behavior considering thickness variation, dome curvature, and internal pressure; explained stability and deformation using mechanics principles
- Compared two-piece and three-piece can manufacturing; evaluated production processes for material efficiency and waste reduction
- Evaluated sustainability and human factors by linking can dimensions and material use to energy consumption and ergonomics

Sustainable Shopping Cart (Product Design) | Individual Project

Jul 2020 – Apr 2021 | GEMS Wellington

- Designed a collapsible shopping cart to reduce disposable bag usage and improve portability
- Analyzed ergonomics, weight distribution, and folding mechanisms to inform design decisions
- Modeled in Fusion 360; fabricated FDM and timber prototypes
- Produced engineering drawings and Life Cycle Assessment to evaluate environmental impact

TECHNICAL SKILLS

CAD & Design: Fusion 360, SolidWorks, SketchUp, engineering drawings

Programming: Python, MATLAB

Prototyping: FDM 3D printing, CNC, hand tools

INTERNSHIPS

VHM Hospital | Biomedical Research Intern

Jul 18, 2022 – Jul 28, 2022 | India

Observed the use of biomedical devices during procedures, assessed device functionality and efficiency, and compiled observations and areas for potential improvement into a report.

VOLUNTEERING

Claylab Education Foundation | Course Designer

Jun 2022 – Jul 2022 | India

Collaborated with a team to design a comprehensive course syllabus; improved mentor assessment techniques and student comprehension methods; enhanced overall course effectiveness and learning experience.

Claylab Education Foundation | Mentorship Program Coordinator

Feb 2023 – Apr 2023 | Virtual

Managed 20 mentor-mentee pairs in a virtual environment; guided pairs to meet weekly targets; conducted bi-weekly mentor meetings.

AWARDS

Top 5 Finalist, QUEERI Sustainable Development Competition, 2020